Wafer Map Anomaly Detection on WM-811K

From Reconstruction Baselines to Pretrained Local Feature Modelling

50.039 Deep Learning · Y2026 · Group 08

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The Problem

Semiconductor fabs generate hundreds of thousands of wafer maps. Most are normal. A few are not.

- **Defects are rare and costly:** each bad wafer drives up yield loss
- **Defect types are unpredictable:** 9 known patterns, but new ones can emerge
- **Labelled defect data is scarce:** 3.1% of the dataset is labelled defective; the rest is unlabelled
- **Classification does not scale:** you cannot enumerate every failure mode at training time

> **Goal:** Learn what a normal wafer looks like. Flag anything that deviates.

The Dataset: WM-811K

811,457 wafer maps from the LSWMD dataset

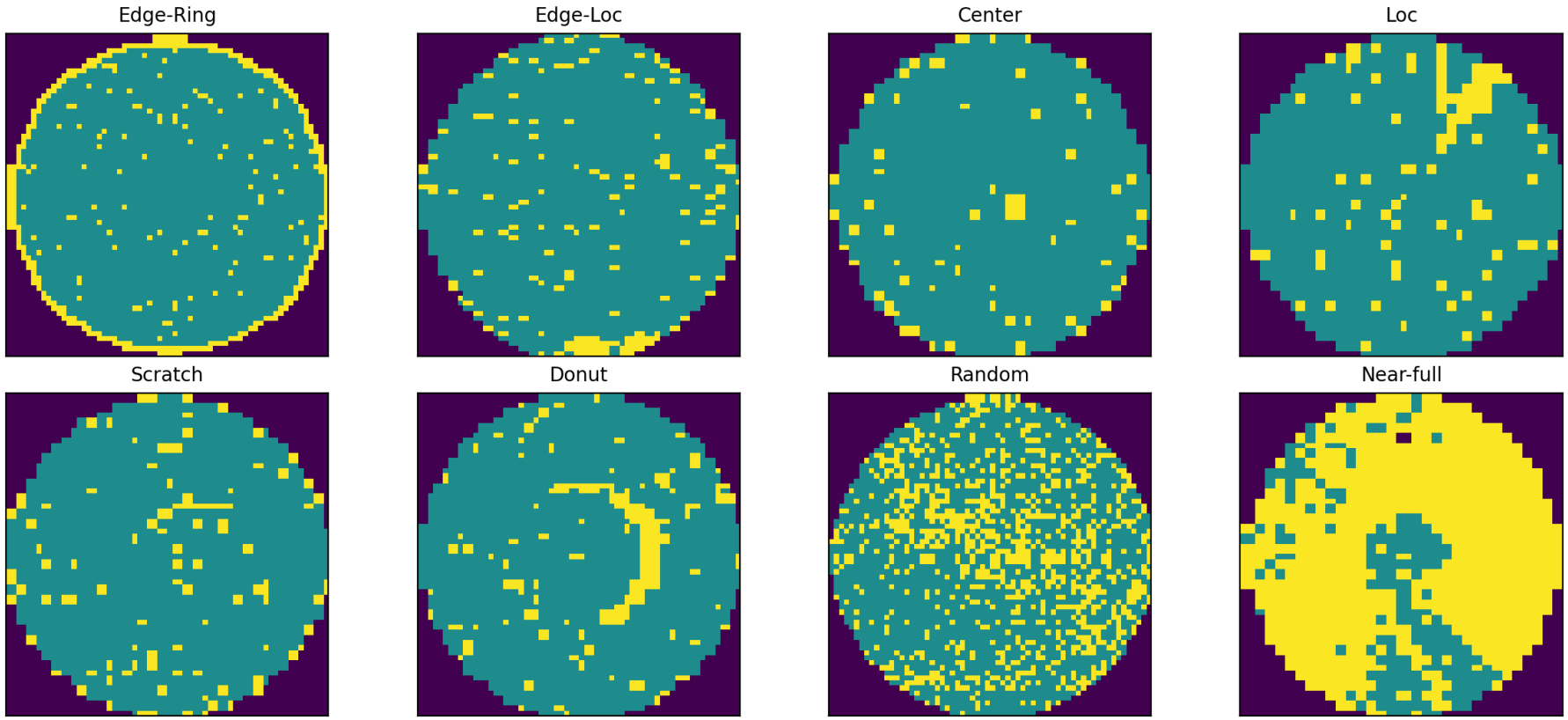
Split	Count	Share
Normal (labelled)	147,431	18.2%
Defective (labelled)	25,118	3.1%
Unlabelled	638,908	78.7%

Key challenge: Class imbalance: a model predicting every wafer as normal achieves 96%+ accuracy while catching zero defects.

The 9 defect pattern types differ in shape, size, and sparsity, making detection non-trivial.

Defect Type Gallery

Representative Defect Patterns from WM-811K Test Set



Defect Types: A Spatial Challenge

Eight distinct defect morphologies spanning very different spatial scales.

Defect type	Count (test)	Spatial character
Edge-Ring	84	Continuous ring of failures near the wafer perimeter
Edge-Loc	53	Localised cluster at or near the wafer edge
Center	50	Compact cluster at the wafer centre
Loc	34	Off-centre localised patch
Scratch	15	Thin elongated streak across the die grid
Donut	7	Curved partial ring inside the wafer body
Random	5	Scattered failures with no spatial coherence
Near-full	2	Defective region covering nearly the entire wafer

Key insight: Broad defects (Edge-Ring, Center) are detectable by any global method. Small, sparse defects (Scratch, Loc) require spatially precise, **patch-level** scoring.

Our Plan: Normal-Only Training

We train **exclusively on normal wafer maps**. At inference, each wafer receives an anomaly score (higher means more deviant from normal).

Why not supervised classification?

- Requires balanced, labelled defect data: scarce in production
- Cannot detect defect types unseen at training time
- Anomaly detection generalises to new defect patterns automatically

Training plan:

- **Train:** normal wafers only
- **Validation:** normal wafers only (threshold calibration)
- **Test:** normal + all labelled defect types (held out completely)

Benchmark Split & Threshold Protocol

All experiments share a **single fixed split** drawn from 50,000 labelled-normal wafers.

Data split

Subset	Normal	Defect
Train	40,000	0
Val	5,000	0
Test	5,000	250

Defects are capped at **5% of test-normal count** to reflect realistic production prevalence.

Threshold rule

$\tau_{0.95}$ = 95th percentile of **validation-normal** anomaly scores only

- No defect labels used at any stage
- Exactly 5% of validation normals exceed $\tau_{0.95}$ by construction
- Bounding expected false-positive rate at ≈5%

> **Best-sweep F1** (threshold tuned on test labels) is reported as an upper bound only. It is never used as the deployment metric.

How We Evaluate

Three metrics, because **accuracy alone is useless** for imbalanced data.

Metric	What it measures	Why it matters here
AUROC	Ranking quality across all thresholds	Threshold-free; strong ranking signal
AUPRC	Precision-recall area under curve	**Our primary metric** -- most informative at 5% prevalence
F1	Harmonic mean at the deployed threshold	Reflects real production performance

A model predicting all wafers as normal achieves **96%+ accuracy** and **zero defect recall**. AUROC and AUPRC expose this immediately.

Threshold: 95th percentile of normal-only validation scores, no defect labels required.

Method Journey

Each method family was selected to address the limitations of its predecessor.

	Family	Hypothesis	Finding
H1	Reconstruction	Can pixel reconstruction detect anomalies?	Pixel errors too diffuse; sparse defects invisible
H2	Global Features	Do pretrained global features do better?	Pooling collapses spatial structure
H3	**PatchCore + CNN**	**Do local patch features matter?**	**Large jump: patch-level scoring is the key**
H4	PatchCore + ViT	Does backbone architecture matter?	ViT-B/16 at 224×224 wins
H5	PatchCore + ViT (MAE)	Does domain adaptation improve features?	AUPRC improves; F1 stays stable
H6	**Ensemble**	**Can diverse pretraining push further?**	**Best overall: AUROC 0.967**

Phase 1: Reconstruction Baselines

The autoencoder compresses normal wafers into a 128-dim bottleneck. Anomalies produce elevated reconstruction error.

Architecture

- Encoder: 3× strided Conv2d (16→32→64 ch, stride=2) → flatten → 128-dim latent
- Decoder: symmetric transposed convolutions → sigmoid output
- 99.6% compression (4096 pixels → 128 values)

Variants tested (11 AE runs + VAE + SVDD)

- BatchNorm, Dropout sweep {0.00, 0.05, 0.10, 0.20}
- Residual skip connections
- Resolutions: 64×64, 128×128, 224×224
- VAE (β ∈ {0.001, 0.005, 0.01, 0.05}) and Deep SVDD

Key Phase 1 results

Variant	F1	AUROC	AUPRC
AE 224×224 (topk)	**0.510**	**0.901**	**0.596**
AE+BN 64×64 (max_abs)	0.502	0.834	0.568
AE baseline 64×64	0.468	0.839	0.522
VAE (β=0.005)	0.340	0.772	0.372
Deep SVDD	0.360	0.788	0.213

Resolution beats architecture: 224×224 plain AE (F1=0.510) outperforms all 64×64 variants including BatchNorm, dropout, and residual AEs.

Phase 1: The Score Ablation Insight

The scoring rule matters almost as much as the model. Seven reduction rules tested on the same fixed AE checkpoint:

Scoring Rule	Definition	F1
topk_abs_mean	Average top 20% of pixel errors (sorted by magnitude)	**0.468**
mse_mean	Average squared pixel errors across entire image	0.410
max_abs	Single worst (maximum absolute) pixel error	0.324
foreground_mse	Average error in foreground-masked region	0.317
mae_mean	Average absolute pixel errors	0.300
pooled_mae_mean	MAE with spatial pooling	0.293
foreground_mae	MAE in foreground region	0.239

topk_abs_mean wins by +0.057 F1 over mse_mean. Selecting high-error pixels outperforms averaging all pixels.

> The BatchNorm AE rescored with max_abs improved from F1=0.431 → **0.502 without any retraining**. Score design is a first-class decision co-equal with model architecture.

Phase 1: The Global-Averaging Bottleneck

Why reconstruction fails on small defects: a simple arithmetic argument.

A 50-pixel Scratch defect in a 4096-pixel image:

```
wafer score = (50 high-error pixels + 3946 low-error pixels) / 4096  ≈  weak signal
```

The defect is **arithmetically diluted** even though local error is clearly elevated.

Per-defect recall (baseline AE)

Defect	Recall
Edge-Ring	**0.810**
Center	**0.720**
Loc	0.206
Scratch	0.200

Large defects survive averaging; small ones drown.

Phase 1 ceiling: AUPRC=0.596, F1=0.510

Any improvement beyond this must fundamentally change how anomaly signals are aggregated: from global, full-image average to **local, patch-level scores**.

> Resolution helps (224×224 > 64×64), but the Scratch/Loc problem remains. Native-resolution autoencoders still score globally. Small defects still drown.

Phase 2: Pretrained Features (Negative Control)

Can richer ImageNet features solve the problem without local scoring?

Setup: Frozen ResNet18 and WideResNet50-2. Score = L2 distance from each test wafer's global embedding to the centroid of train-normal embeddings. No training required.

Backbone	F1	AUROC	AUPRC
WideResNet50-2	0.243	0.677	0.142
ResNet18	0.236	0.685	0.195
(AE baseline, for ref)	*0.468*	*0.839*	*0.522*

Both baselines fall **below every reconstruction model** despite far richer features (512-dim vs 2048-dim).

Why capacity does not help

Spatial mean-pooling collapses the entire feature map into a single global vector:

- A localised defect contributes **equally** to the global vector as background noise
- WideResNet50-2 (2048-dim) fails for the same reason as ResNet18 (512-dim)
- **The bottleneck is aggregation, not feature quality**

> This is a deliberate negative control. It proves the problem is spatial, not representational, and motivates patch-level scoring in Phase 3.

Phase 3: Local Spatial Scoring (Three Methods)

Teacher-Student Distillation

Train a lightweight student CNN to reproduce the frozen teacher's spatial feature maps on normal wafers. Anomaly score = per-location mismatch.

Backbone	F1	AUROC
ResNet50	**0.525**	0.909
WRN50-2 multilayer	0.524	**0.923**
WRN50-2 layer2	0.508	0.920
ResNet18	0.495	0.894
ViT-B/16	0.163	0.661

ViT-B/16 catastrophically fails: CNNs cannot mimic globally context-aware ViT tokens via locally-growing receptive fields → architectural mismatch.

FastFlow

Normalising flow on WideResNet50-2 features: F1=0.482, AUROC=0.871. Strong on Loc (recall 0.588), weak on Scratch (recall 0.133).

PatchCore (training-free)

Store coreset-reduced memory bank of patch embeddings from normal wafers. Score = nearest-neighbour distance. No training required.

Backbone	F1	AUROC	AUPRC
WRN50-2 64×64	0.532	0.917	0.562
EffNet-B0 224×224	0.544	0.925	0.483
WRN50-2 224×224	0.549	0.931	0.659
EffNet-B1 240×240	0.591	0.935	0.609
ViT-B/16 224×224	**0.595**	**0.956**	**0.671**

> PatchCore with no training beats every teacher-student and FastFlow variant. Phase 3 ceiling: F1=0.595 (PatchCore + ViT-B/16).

Phase 3: What Drove the PatchCore Gains?

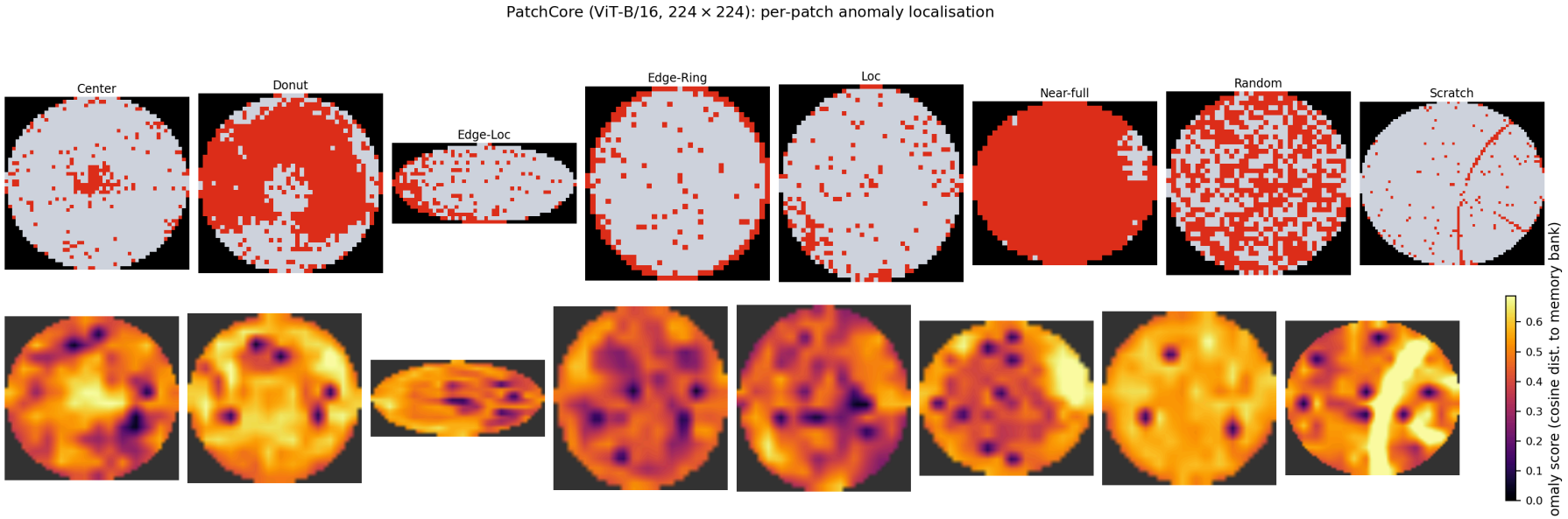
Three independent factors, each contributing a measurable, attributable lift.

Factor	Change	F1 gain
In-domain → ImageNet pretrained	AE-BN encoder → ResNet18	+0.065
Backbone capacity	ResNet18 → WideResNet50-2 multilayer at 64×64	+0.131
Native resolution preprocessing	EfficientNet-B0: 64×64 → 224×224	+0.077
CNN → ViT at same resolution	EfficientNet-B0 x224 → ViT-B/16 x224	+0.004

Backbone capacity is the single largest lever (+0.131). Resolution is the second-largest and fully independent (+0.077). The ViT advantage over CNN is real but small in isolation. Its main contribution is concentrated on the hardest defect classes.

> **Why resolution matters so much for ViT:** At 64×64, ViT-B/16's fixed 16×16 patch size yields only **16 tokens** instead of 196. Each token covers 4× more spatial area in each dimension. Defects spanning fewer pixels than one token become completely invisible to the memory bank. F1 collapses from 0.595 (224×224) to 0.342 (64×64), a drop of −0.253.

PatchCore: Local Patch Scoring in Action



Top row: original wafer binary maps (red = failed dies). Bottom row: per-patch anomaly score heatmap at the ViT-B/16 patch grid resolution. Patch-level scoring localises small defects that pixel-averaged methods miss entirely.

ViT-B/16 PatchCore: Architecture & Block Depth

Pipeline: 224×224 input → 14×14 grid = **196 non-overlapping 16×16 patches** → 768-dim tokens → 12 Transformer blocks (12-head self-attention, dim=768) → block 6 patch embeddings → coreset memory bank (5% subsampling ≈400k patches) → **max cosine distance** over all 196 patches.

Each token has **global receptive field from block 1**, unlike CNNs where receptive fields grow gradually through strided convolutions.

Block depth sweep (all other hyperparameters fixed):

Block	Depth	F1	AUROC	AUPRC	Recall
3	Early	0.550	0.934	0.622	73.2%
6	**Mid**	**0.580**	**0.954**	0.642	**80.0%**
9	Mid-late	0.580	0.943	**0.677**	78.8%
11	Final	0.435	0.866	0.490	56.4%

> **Block 11 collapses (F1=0.435):** by the final block, self-attention has pooled all representations into globally class-discriminative embeddings that **lose the fine-grained spatial diversity** PatchCore requires. This is consistent with DINO findings that intermediate ViT layers retain the strongest local structure for dense prediction.

Phase 4: MAE Fine-Tuning for Domain Adaptation

Motivation: Standard PatchCore uses frozen ImageNet weights that have never seen wafer maps. Can masked patch reconstruction force it to learn wafer-specific local statistics?

Setup: Fine-tune the last 6 ViT blocks (6–11) on 40k training normals. Decoder (2-layer MLP) discarded after training. AdamW, LR=1e-4, 10 epochs.

Results

Backbone	F1	AUROC	AUPRC	Recall
Frozen ImageNet	0.595	0.956	0.671	~80%
MAE 75% mask	0.595	0.959	0.717	82.4%
MAE 25% mask	**0.594**	**0.962**	**0.762**	**84.4%**

Deployed F1 is **stable**. AUPRC improves by **+0.091** (25% mask vs frozen).

Why 25% masking outperforms 75% on wafer maps

Natural images have dense, spatially redundant texture, so high masking is hard and informative.

Wafer maps are **sparse binary grids** with far less texture diversity:

- At 75% mask → only 49 of 196 patches visible; insufficient local context
- At 25% mask → 147 patches visible; richer context per training step

Effect on hardest class: Scratch recall: 75% mask = 45.5% → 25% mask = **63.6%**

> Masking ratio is a meaningful hyperparameter for sparse industrial images. The natural-image default of 75% is **suboptimal** for this domain.

Top 10 Models

64 training runs: Reconstruction: AE (11) · VAE (9) · SVDD (1) · Center-distance (2) · Teacher-Student (5) · FastFlow (3) · RD4AD (1) · PatchCore: CNN (14) · ViT (8) · DINOv2 (3) · Ensemble (5) · Supervised (1)

Rank	Family	Model	Sweep F1	AUROC	AUPRC
1	Ensemble	**ViT-B/16 + DINOv2 (max-fusion)**	**0.676**	**0.967**	0.716
2	PatchCore + ViT	ViT-B/16, MAE fine-tuned (25% mask)	0.692	0.962	**0.762**
3	PatchCore + ViT	ViT-B/16, MAE fine-tuned (75% mask)	0.668	0.959	0.717
4	PatchCore + ViT	ViT-B/16, frozen, 224×224	0.651	0.956	0.671
5	PatchCore + CNN	EfficientNet-B1, 240×240	0.653	0.935	0.609
6	PatchCore + CNN	WideResNet50-2, 224×224	0.634	0.931	0.659
7	Teacher-Student	ResNet50	0.606	0.909	0.599
8	Teacher-Student	WideResNet50-2	0.561	0.923	0.546
9	PatchCore + ViT	DINOv2 ViT-B/14, block-9	0.570	0.915	0.561
10	Reconstruction	Autoencoder, 224×224	0.587	0.901	0.596

Best Model: ViT-B/16 + DINOv2 Ensemble

Per-defect recall

Defect	ViT-B/16	DINOv2	Ensemble
Scratch	0.67	**0.93**	**0.80**
Loc	**0.79**	0.71	**0.91**
Edge-Loc	**0.79**	0.55	**0.85**
Center	**0.82**	0.56	**0.88**
Edge-Ring	**0.89**	0.74	**0.89**
Donut	1.00	1.00	**1.00**

Where one model is weak, the other is strong.

Overall metrics

Model	AUROC	AUPRC	F1
ViT-B/16	0.956	0.671	0.595
DINOv2	0.915	0.561	0.492
Ensemble (max-fusion)	0.967	0.716	**0.623**
Ensemble (Mahalanobis)	**0.968**	**0.762**	0.592

> Mahalanobis requires joint signal from both models. Better ranking (AUROC, AUPRC) but stricter threshold lowers deployed F1.

ViT-B/16 vs DINOv2: Two Different Views of the Same Image

	ViT-B/16	DINOv2 ViT-B/14
Pretraining	Supervised ImageNet classification	Self-supervised self-distillation (no labels)
Objective	Predict the correct object class	Make semantically similar images cluster together
What it learns	Discriminative features that separate categories	Semantic features that capture visual similarity
Sensitivity	Structured, boundary-anchored defects	Texture and sparse local pattern deviations
Patch size	16×16 px (196 tokens per image)	14×14 px (256 tokens, finer granularity)
Scratch recall	0.67 (weaker on thin linear defects)	**0.93** (stronger on sparse, low-coverage patterns)
Loc recall	**0.79**	0.71

Because the two models fail on different defect types, taking the maximum anomaly score per wafer recovers detections that either model alone would miss.

Why Ensembling Only Works With Complementary Pretraining

Not all ensembles help. The ViT+EfficientNet-B1 ensemble **degrades** performance.

Ensemble results (main benchmark)

Ensemble	F1	AUROC	AUPRC
ViT-B/16 + DINOv2 (max)	**0.623**	**0.967**	**0.716**
ViT-B/16 alone	0.595	0.956	0.671
ViT-B/16 + EfficientNet-B1 (80/20)	0.576	0.952	0.643
MAE ViT (25%) + DINOv2 (max)	0.544	0.941	0.693

Why ViT+EfficientNet fails

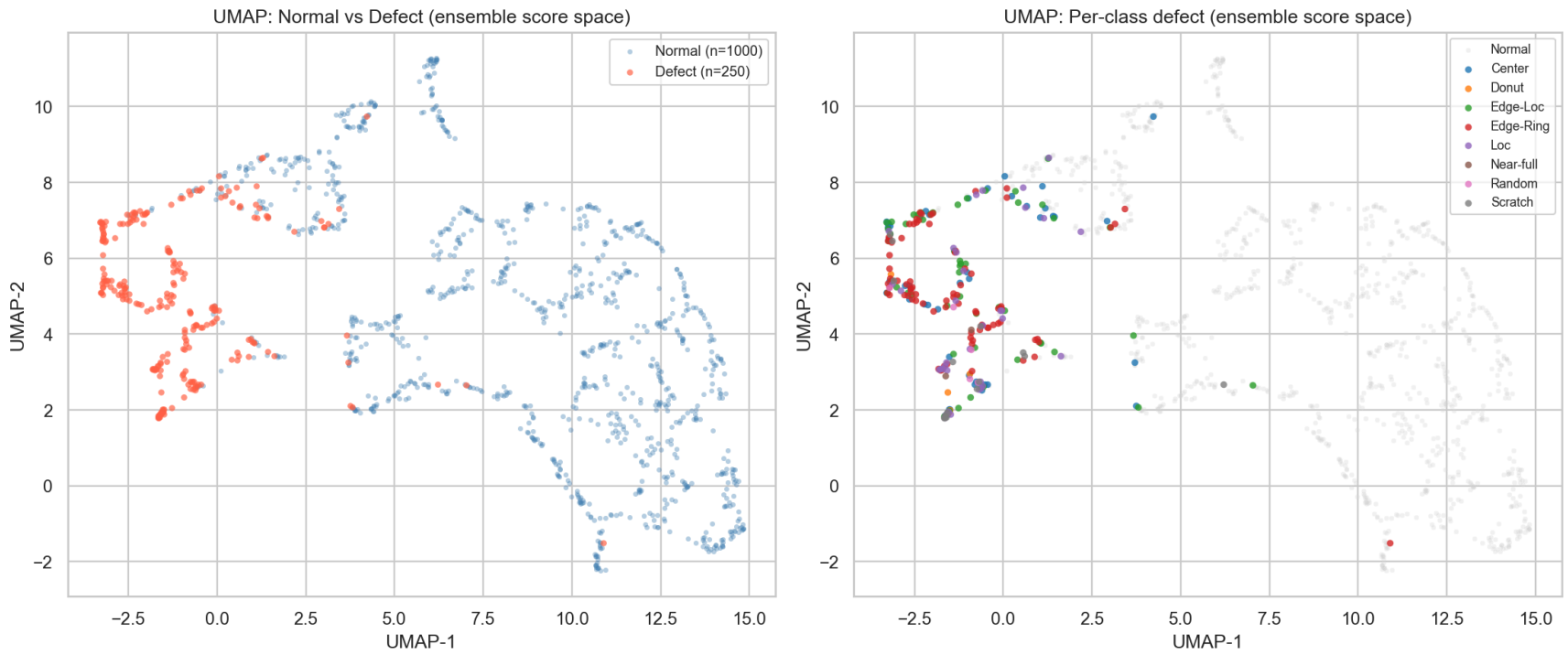
Both trained on supervised ImageNet classification → highly correlated anomaly scores → max-fusion adds no new signal.

Why MAE ViT+DINOv2 also fails

MAE fine-tuning pulls ViT representations toward the same wafer domain as DINOv2's self-distillation objective. The models then **agree on the same wafers**, losing the diversity that enables ensemble gains. The two models' near-zero correlation ($r = -0.016$) confirms they capture largely independent signals in the **frozen** configuration.

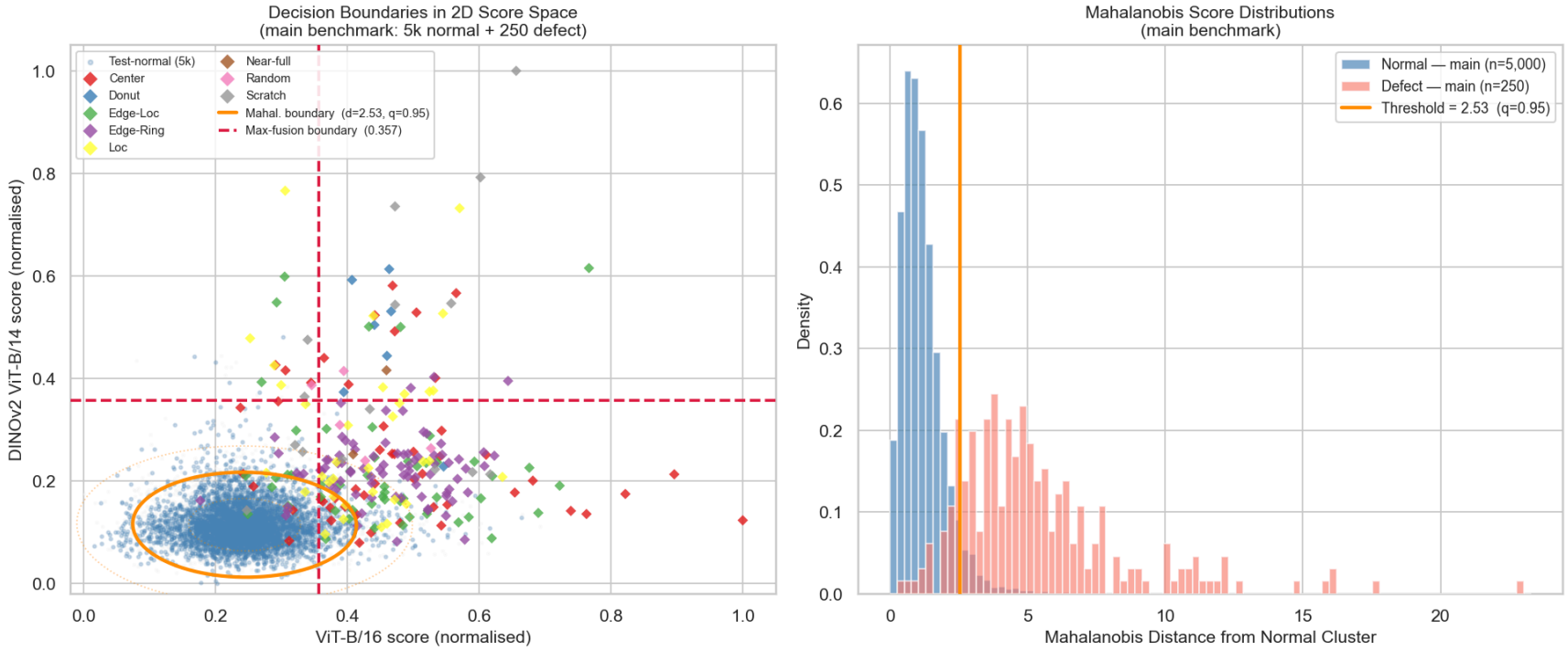
> **Design principle:** Ensemble gains require genuinely different objectives, not just different architectures or backbones.

Complementary Signals in Score Space



UMAP of [ViT score, DINOv2 score] per test wafer. Left: normal vs defect separation. Right: per-class clusters. Scratch concentrates where DINOv2 is high; Edge-Ring and Edge-Loc concentrate where ViT is high. Max-fusion captures both regions.

Mahalanobis Fusion: Tighter Decision Boundary in Score Space



Left: Mahalanobis ellipse (orange) vs max-fusion L-shape (red dashed) in normalised score space. Mahalanobis requires joint signal from both models. Right: score distributions show clear separation. Mahalanobis achieves AUROC 0.968, AUPRC 0.762, Sweep F1 0.722, all above max-fusion.

Expanded Holdout: Does It Generalise?

A 14× larger disjoint evaluation pool: **70,000 normal + 3,500 defect wafers** (same train/val, new test set).

Model	F1	AUROC	AUPRC	Sweep F1
PatchCore + EfficientNet-B1 240×240	**0.596**	0.953	0.656	0.671
PatchCore + ViT-B/16, MAE 25%	0.585	0.958	0.724	0.697
Ensemble: ViT + DINOv2 (Mahalanobis)	0.573	**0.963**	**0.731**	**0.702**
Ensemble: ViT + DINOv2 (max-fusion)	0.567	0.953	0.645	0.628
PatchCore + ViT-B/16 (frozen)	0.548	0.941	0.615	0.606

Key reversals vs main benchmark:

- **EfficientNet-B1 overtakes ViT-B/16:** its compact UMAP cluster geometry is more stable at scale
- **Max-fusion drops F1=0.623 → 0.567:** precision loss at scale. Max-fusion behaves as a logical OR, firing on one-sided outliers
- **Autoencoder excluded:** 90% of 70k holdout normals exceed $\tau_{0.95}$ calibrated on 5k validation normals, a concrete illustration of threshold instability

Thresholding Limitations

The 95th-percentile threshold is fair and label-free, but a consistent gap to best-sweep F1 remains.

Score-distribution overlap is the root cause

Even the strongest models show substantial overlap between normal and anomaly distributions near the operating threshold. Low-scoring defects fall below $\tau_{0.95}$ regardless of where the threshold sits.

Model	Deployed F1	Best-sweep F1	Gap
AE-BN	0.431	0.630	0.199
WRN50-2 PatchCore x224	0.549	0.634	0.085
ViT-B/16 PatchCore	0.595	0.651	0.056

Moving to the best-sweep threshold raises the cut-point (stricter), improving precision sharply but at the cost of recall. It is a precision/recall trade, not a free lunch.

Why the gap persists (structural, not fixable by better models alone)

Defects are sparse and small, producing only a **handful of anomalous patches per wafer**. The wafer-level score (max over 196 patches) cannot deviate far from the normal distribution when only 2–5 patches are anomalous.

Three viable improvement directions:

1. **Manifold-aware calibration:** anomalies at manifold boundaries are farther from normal neighbours; use this geometry to derive the threshold
2. **Parametric tail modelling:** fit the validation-normal score tail and derive a principled cutpoint
3. **Small-defect calibration:** even a few confirmed defect scores at deployment could inform a better operating point without contaminating training

What If We Had Used Defect Labels?

Our entire approach assumes **normal-only training**. But what if that assumption costs us performance?

Two experiments to test this:

- **Defect-tuning PatchCore:** Fine-tune ViT-B/16 backbone layers with contrastive loss on defect examples
- **Supervised linear probe:** Train a classifier on up to 1,016 labelled defect samples across all defect classes, with **Scratch and Loc withheld**, to test generalisation

> **The question:** Does knowing what defects look like actually help?

Result: Normal-Only Is the More Robust Choice

Supervised linear probe sweep (Scratch + Loc withheld from training):

Defect %	Train defects	AUROC	F1	Sweep F1	Scratch recall	Loc recall
0.1%	20	0.895	0.499	0.640	0.083	0.400
1.0%	203	0.941	0.554	0.721	0.333	0.657
5.0%	1,016	0.956	0.581	0.760	0.167	0.686
PatchCore (no labels)	**0**	**0.956**	**0.595**	**0.651**	**0.727**	**0.829**

With 1,016 labels, supervised AUROC matches PatchCore. But Scratch recall is **0.167** vs **0.727**: the classifier never saw Scratch during training.

Why Scratch doesn't improve: In the ViT-B/16 feature space, Scratch sits geometrically adjacent to the normal wafer mass, not to other defect clusters. No amount of seen-class supervision shifts that boundary.

> **Conclusion:** Anomaly detection has no blind spots. It flags anything that looks unusual, whether or not that defect type existed at training time.

Future Work

Three directions with the clearest path to measurable gains.

Better thresholding

The largest gap between deployed F1 and best-sweep F1 comes from a fixed percentile threshold, not from weak features.

- **Manifold-aware calibration:** wafers near the boundary of the normal manifold have larger distances to their nearest neighbours; use this geometry to set a tighter, per-region threshold
- **Parametric tail modelling:** fit an extreme-value distribution to validation-normal scores and derive a principled cut-point instead of a percentile heuristic

Domain-adapted features

MAE fine-tuning on 40k normals already improved AUPRC by +0.091. Further gains are plausible with:

- **Larger fine-tuning sets:** the unlabelled pool has 638k wafers
- **Contrastive or reconstruction objectives on raw wafer data:** reduce the gap between ImageNet texture features and binary die-grid statistics

Better ensemble design

Max-fusion degrades at scale (F1 drops 0.623 to 0.567 on the 70k holdout). A learned, score-space meta-learner trained on validation normals alone could improve precision without needing defect labels.

Conclusion

Wafer map anomaly detection is a spatial problem, and the solution must be spatial.

- **Reconstruction baselines plateau at AUPRC 0.596.** Global averaging dilutes small defect signals regardless of architecture or capacity.
- **PatchCore with ViT-B/16 reaches AUPRC 0.671** by scoring each patch independently. Patch-level scoring is the single most impactful design decision.
- **MAE fine-tuning on wafer data pushes AUPRC to 0.762** with no label cost. The masking ratio matters: 25% beats 75% on sparse binary grids.
- **Ensembling complementary pretraining objectives reaches AUROC 0.968.** ViT-B/16 (supervised) and DINOv2 (self-supervised) fail on different defect types; each fusion method gives the better score on different metrics.
- **Normal-only training generalises. Supervised classifiers do not.** With 1,016 labels, a linear probe matches PatchCore AUROC but misses withheld defect types entirely.

> The key insight is that a model need not know what a defect looks like. It only needs to know what normal looks like well enough to notice when something is not.